Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)		Study of {space / stars / galaxies} (1) don't accept planets For the 2nd mark either: At {different / various} {wavelengths / frequencies / energies} of the em spectrum **OR** Of various regions of the em spectrum	2			2		
	(b)		W in base units → kg m^2 s^{-3} **or** J → kg m^2 s^{-2} **or** N → kg m s^{-2} (1) Units of σ = kg s^{-3} K^{-4} (1)	1	1		2	2	
	(c)	(i)	Use of $T = \frac{W}{\lambda_{max}}$ (re-arrangement and substitution) i.e. $T = \frac{2.9 \times 10^{-3}}{674 \times 10^{-9}}$ (1) T = 4303 K (1) **unit mark**		2		2	2	
		(ii)	Red / orange	1			1		
	(d)		Correct substitution into $P = A\sigma T^4$ i.e. $P = 5.67 \times 10^{-8} \times 4\pi \times (1.39 \times 10^{10})^2 \times (4303)^4$ **ecf** on T **Or** $P = 4.72 \times 10^{28}$ [W] seen (1) Recall $I = \frac{P}{4\pi D^2}$ (1) **Alternative 1** - Substitution and rearrange i.e. $D = \sqrt{\frac{4.72 \times 10^{28}\,\textbf{ecf}}{4\pi \times 3.09 \times 10^{-8}}}$ (1) $D = 3.49 \times 10^{17}$ [m] (1) D in light years = $\frac{3.49 \times 10^{17}}{9.46 \times 10^{15}}$ = 36.8[5 ly] so claim correct **ecf** on D (1)			5	5	5	